

SELF-INDUCTION AND THE PROTECTIVE TILT: A HEAT-KERNEL MONOTONICITY FOR VORTICES IN ASYMMETRIC STRAIN

ADILKHAN SALKIMBAYEV

ABSTRACT. The velocity a vortex tube feels splits, through the Biot–Savart law, into the field of the other vortices and its own field. In the exact two-dimensional reduction of a tube in a uniform strain (the asymmetric Burgers problem), the first tears the vortex and the second holds it together. We make the second precise. We construct a circulation weighted by a backward heat kernel adapted to the strain, for which the strain and the viscosity cancel exactly; its rate of change is the self-induced term alone. That term has the sign of a single geometric quantity, the tilt of the vortex relative to the strain axes, and by exact moment identities it is non-negative precisely when the vortex is rounder than the passive field that would occupy its place. We prove that the tilt has the protective sign: for weak vortices at every asymmetry $\lambda < 1$; to first order in λ at every circulation, by an energy argument built on the skew-symmetry of the linearised Biot–Savart operator found by Galloway and Wayne; and for large circulation at every $\lambda < 1$, where the protection also holds for the full Gaussian weight. We prove the time-dependent version at first order in λ for early and late times. Numerically, every sign also holds at intermediate circulation. In the torn regime $\lambda > 1$, where no steady vortex exists, the lifetime gained over the passive field collapses onto a single curve of the scale-invariant ratio $\varepsilon = 2\pi\lambda\nu/\Gamma$, with a threshold near $\varepsilon_c \approx 0.05$. We discuss what this implies, and what it does not, for a collapsing tube in three dimensions.

1. INTRODUCTION

A vortex tube in a turbulent flow is stretched along its axis and strained across it by the velocity that the rest of the vorticity induces. The axial part concentrates the tube; the in-plane part, once it has an extensional direction, pulls it apart. The tube also moves in its own field, and its own rotation competes with the in-plane strain. Burgers’ vortex [1] is the exact balance in axisymmetric strain. In asymmetric strain the vortex is no longer explicit; its existence and structure were studied numerically by Robinson and Saffman [13] and by Prochazka and Pullin [12], asymptotically by Moffatt, Kida and Ohkitani [11], and rigorously by Galloway and Wayne [5, 4] and by Maekawa [8, 9]; see the survey [3].

This paper isolates the part of the Biot–Savart law that protects the vortex and shows that it has a definite sign. The motivation is the search, recorded in earlier papers of this series [14, 15], for a scale-invariant quantity with a monotonicity property, of the kind Huisken’s weighted area provides for mean curvature flow [6]. Such formulas use a backward heat kernel to cancel the linear part of the dynamics and leave a term of definite sign. Here the linear part is strain plus viscosity, and the term that remains is self-induction.

Results. Write λ for the asymmetry of the strain and $R = \Gamma/\nu$ for the circulation Reynolds number (Section 2).

- (Proposition 2.3.) For a Gaussian weight ϕ evolving backward under the adjoint of strain plus viscosity, $\frac{d}{dt} \int \omega \phi = \int \omega v \cdot \nabla \phi$, where v is the self-induced velocity.

- (Propositions 2.4 and 2.2.) The quadratic part of this term is $-\frac{c(q-p)}{2\pi}T$, with $q \geq p$ and T the *tilt*. By exact moment identities, $T \leq 0$ if and only if the vortex is rounder than its passive twin.
- (Theorem 3.1.) For weak vortices and every $\lambda \in (0, 1)$: $T < 0$.
- (Theorem 4.2.) To first order in λ , for every circulation: $T/\lambda \leq -\frac{16\pi^3}{\Gamma}\|V\|^2 < 0$. The proof uses only that diffusion with strain is self-adjoint and dissipative, and that the linearised Biot–Savart operator is skew-symmetric in Galloway and Wayne’s Gaussian-weighted space.
- (Theorems 5.1 and 5.2.) For every $\lambda < 1$ and large circulation: $T = -4\pi\Gamma\lambda + O(1)$, and $\int \omega v \cdot \nabla \phi > 0$ for the *full* Gaussian weight whenever it is wider along the compressed axis.
- (Proposition 4.3.) Time-dependent version from a round start, at first order in λ , for $t \lesssim 1/R$ and for $t \gtrsim 2 \ln R$.
- (Section 6.) Numerical evidence in the remaining ranges, and the threshold law $\varepsilon_c \approx 0.05$ for the lifetime of a vortex in the torn regime.

2. SETTING AND EXACT IDENTITIES

Let the strain be $(\gamma_1 x_1, \gamma_2 x_2, \gamma_3 x_3)$ with $\gamma_{1,2} = -\frac{\gamma}{2}(1 \pm \lambda)$ and $\gamma_3 = \gamma > 0$. A tube along x_3 with vorticity $\omega(x_1, x_2, t)$ is an exact solution of the Navier–Stokes equations if ω solves the reduced equation below [5]. In units $\nu = \gamma = 1$,

$$\partial_t \omega + \nabla \cdot ((u + v)\omega) = \Delta \omega, \quad u = -(g_1 x_1, g_2 x_2), \quad g_{1,2} = \frac{1}{2}(1 \pm \lambda), \quad v = K * \omega, \quad (1)$$

with $K(x) = x^\perp / (2\pi|x|^2)$. Equivalently $\partial_t \omega + v \cdot \nabla \omega = (\mathcal{L} + \lambda \mathcal{M})\omega$, with $\mathcal{L} = \Delta + \frac{1}{2}x \cdot \nabla + 1$ and $\mathcal{M} = \frac{1}{2}(x_1 \partial_1 - x_2 \partial_2)$. The circulation $\Gamma = \int \omega$ is conserved, and $R = \Gamma/\nu = \Gamma$ in these units. For $\lambda < 1$ both in-plane rates are compressive; for $\lambda > 1$ the direction x_2 is extensional. The *passive twin* is the solution of (1) with $v \equiv 0$. Its steady state is the Gaussian ΓG_λ with variances $1/g_1$ and $1/g_2$. We write $G = G_0 = \frac{1}{4\pi}e^{-|x|^2/4}$.

Definition 2.1. The *tilt* is $T = \iint \omega(x)\omega(y)h(x-y)dx dy$, where $h(r) = r_1 r_2 / |r|^2$. Also let $P = \iint \omega(x)\omega(y)\frac{r_1^2 - r_2^2}{|r|^2}$, and $M_{ij} = \Gamma^{-1} \int \omega x_i x_j$.

For an ellipse-like vortex, $T < 0$ means that its long axis lies between the extensional axis x_2 and the compressive axis x_1 , counted in the sense of rotation (for $\Gamma > 0$).

Proposition 2.2 (Moment identities). *For every solution of (1),*

$$\Gamma M'_{11} = -2g_1 \Gamma M_{11} + 2\Gamma - \frac{T}{2\pi}, \quad \Gamma M'_{22} = -2g_2 \Gamma M_{22} + 2\Gamma + \frac{T}{2\pi}, \quad \Gamma M'_{12} = -\Gamma M_{12} + \frac{P}{4\pi}.$$

Consequently, if $D_{11} = M_{11} - M_{11}^{\text{pass}}$ (same initial data), $T = -2\pi\Gamma(D'_{11} + 2g_1 D_{11})$, and at a steady state $T = 4\pi\Gamma(1 - g_1 M_{11})$.

Proof. Pair (1) with $x_i x_j$. The strain and Laplacian terms are elementary. For the self-induced term, $\int \omega x_i v_j = \frac{1}{2} \iint \omega \omega r_i K_j(r)$ by the antisymmetry of K . This gives $\int \omega x_1 v_1 = -T/4\pi$, $\int \omega x_2 v_2 = T/4\pi$, and $\int \omega(x_2 v_1 + x_1 v_2) = P/4\pi$. The passive twin obeys the first identity with $T = 0$. $\square$

At a steady state, therefore, $T < 0$ if and only if $M_{11} > 1/g_1$: the vortex is wider along the compressed axis than its passive twin, i.e. rounder.

Proposition 2.3 (Adjoint weight). *Let ϕ solve $\partial_t \phi + u \cdot \nabla \phi + \Delta \phi = 0$. Then $\frac{d}{dt} \int \omega \phi = \mathcal{S} := \int \omega v \cdot \nabla \phi$. The Gaussians $\phi = c e^{-px_1^2 - qx_2^2}$ solve it if*

$$p' = 2g_1 p + 4p^2, \quad q' = 2g_2 q + 4q^2, \quad c'/c = 2(p + q),$$

and if $p(T_*) = q(T_*) > 0$ at a final time T_* , then $p(t) \leq q(t)$ for $t \leq T_*$.

Proof. $\frac{d}{dt} \int \omega \phi = \int \omega (\partial_t \phi + (u + v) \cdot \nabla \phi + \Delta \phi)$ by (1). The ODEs come from substitution. The ordering follows by comparison, since $g_1 \geq g_2$ and both solutions start from the same final value. $\square$

The strain and the viscosity have dropped out exactly. This is the mechanism of Huisken's formula, with the self-induced velocity in place of the curvature term.

Proposition 2.4 (Quadratic reduction). $\mathcal{S} = \frac{1}{2} \iint \omega(x) \omega(y) K(x - y) \cdot (\nabla \phi(x) - \nabla \phi(y))$. For the quadratic part $\phi_2 = c(1 - px_1^2 - qx_2^2)$ of the weight,

$$\mathcal{S}[\phi_2] = -\frac{c(q - p)}{2\pi} T.$$

Proof. Symmetrise in $x \leftrightarrow y$. For ϕ_2 , $\nabla \phi_2(x) - \nabla \phi_2(y) = -2c \operatorname{diag}(p, q) r$, and $K(r) \cdot \operatorname{diag}(p, q) r = (q - p) r_1 r_2 / 2\pi |r|^2$. $\square$

Remark 2.5. No static positivity holds: an elliptical vortex tilted the other way has $T > 0$ and $\mathcal{S}[\phi_2] < 0$. Positivity is a property of the *dynamics*: it has to be shown that the vortices that actually arise are tilted ahead. Under the reflection $x_2 \mapsto -x_2$, $\omega \mapsto -\omega$ and $\mathcal{S} \mapsto -\mathcal{S}$, so all statements concern $|\int \omega \phi|$; we take $\Gamma > 0$.

Proposition 2.6 (Steady formula). *If ω is a steady solution of (1) and ϕ is a fixed smooth weight of moderate growth, then $\int \omega v \cdot \nabla \phi = -\int \omega \mathcal{L}_\lambda^* \phi$, where $\mathcal{L}_\lambda^* \phi = \Delta \phi - gx \cdot \nabla \phi$.*

Proof. Pair $\Delta \omega - \nabla \cdot ((u + v)\omega) = 0$ with ϕ . $\square$

3. WEAK VORTICES

Fix $\lambda \in (0, 1)$. Gallay and Wayne [4] construct, for $0 < \Gamma \leq \kappa(\lambda)$, a steady solution of (1) that is locally unique in the Gaussian-weighted space $X_\lambda = L^2(G_\lambda^{-1} dx)$. By the same fixed-point argument it has the expansion $\omega = \Gamma G_\lambda + \Gamma^2 w_0 + O(\Gamma^3)$ in X_λ , where

$$L_\lambda w_0 = F := v[G_\lambda] \cdot \nabla G_\lambda, \quad L_\lambda = \Delta + \nabla \cdot (gx \cdot).$$

L_λ is self-adjoint on X_λ (the drift is a gradient), with a spectral gap g_2 on zero-mass functions.

Theorem 3.1. *For each $\lambda \in (0, 1)$ there is $\Gamma_0(\lambda) > 0$ such that, for $0 < \Gamma < \Gamma_0$, the steady vortex satisfies $T = 2\Gamma^3 \langle w_0, \tau \rangle + O(\Gamma^4)$, where $\tau = G_\lambda * h$ and $\langle w_0, \tau \rangle < 0$. In particular $T < 0$, $M_{11} > 1/g_1$, and $\mathcal{S}[\phi_2] > 0$ for every adjoint weight.*

Proof. The tilt is a bounded bilinear form on L^1 , since $|h| \leq \frac{1}{2}$, and $X_\lambda \subset L^1$. $T(G_\lambda, G_\lambda) = 0$ by reflection symmetry. The sign of $\langle w_0, \tau \rangle$ follows from four facts. Write $c_i = 1/g_i$, so $c_1 < c_2$.

(i) *The forcing is a signed quadrupole.* With $\Delta \psi = G_\lambda$, $v = (-\partial_2 \psi, \partial_1 \psi)$, and $|r|^{-2} = \int_0^\infty e^{-s|r|^2} ds$, the Gaussian convolutions factor by axis and give

$$F = x_1 x_2 G_\lambda \frac{c_2 - c_1}{2\pi c_1 c_2} \int_0^\infty \frac{e^{-sx_1^2/(1+2sc_1) - sx_2^2/(1+2sc_2)}}{(1+2sc_1)^{3/2} (1+2sc_2)^{3/2}} ds.$$

This has the sign of $x_1 x_2$. The stream function of an elongated Gaussian is rounder than the Gaussian itself, and the mismatch is a quadrupole.

(ii) w_0 has the sign of $-x_1x_2$. $w_0 = -\int_0^\infty e^{sL_\lambda} F ds$, and e^{sL_λ} is a product of one-dimensional Gaussian kernels $k_s(\xi|\xi_0)$ centred at $e^{-g_s}\xi_0$, with $k_s(\xi|\xi_0) > k_s(\xi|-\xi_0)$ for $\xi, \xi_0 > 0$. So e^{sL_λ} maps functions with the sign of x_1x_2 to functions with the sign of x_1x_2 .

(iii) τ has the sign of x_1x_2 . Pairing r with its reflections, and writing $G_\lambda = \gamma_1(x_1)\gamma_2(x_2)$, $\tau(y) = \int_{r_1, r_2 > 0} h(r)[\gamma_1(y_1+r_1)-\gamma_1(y_1-r_1)][\gamma_2(y_2+r_2)-\gamma_2(y_2-r_2)] dr$. For y in the first quadrant both brackets are negative.

(iv) Hence $\langle w_0, \tau \rangle < 0$, strictly, because $F \not\equiv 0$ for $\lambda > 0$ and all the inequalities are strict on open quadrants. The rest follows from Propositions 2.2 and 2.4. $\square$

Remark 3.2. Formally, the same four facts hold along the time-dependent problem started from a round Gaussian, for *every* $\lambda > 0$, including $\lambda \geq 1$, at leading order in Γ . The passive covariance keeps $c_1(t) \leq c_2(t)$, and the propagator is still a product of reflection-monotone kernels.

4. EVERY CIRCULATION, SLIGHT ASYMMETRY

By [5, Thm. 1.1], for $\lambda \in [0, \lambda_0]$ and all Γ the steady vortex exists, is locally unique, and is smooth in λ . At first order, $\omega = \Gamma G + \lambda \text{Re}[W(r)e^{2i\theta}] + O(\lambda^2)$, where the mode-2 amplitude solves

$$(A - iB)W = \frac{r^2}{4}\Omega, \quad \Omega = \Gamma G, \quad A = \rho^{-1}(\rho \cdot)' + 1 - \frac{4}{r^2}, \quad \rho = re^{r^2/4}, \quad (2)$$

with $BX = \frac{2v_\theta}{r}X + \Omega \Delta_2^{-1}X$, where $v_\theta = \frac{\Gamma}{2\pi r}(1 - e^{-r^2/4})$ and $\Delta_2 = \partial_r^2 + \frac{1}{r}\partial_r - \frac{4}{r^2}$. Here A is $\mathcal{L}$ restricted to mode 2, and $-iB$ is the linearised Biot–Savart operator $-\Gamma\Lambda$ of [5] restricted to mode 2.

Lemma 4.1. *On $L^2(\rho dr)$: (a) A is self-adjoint and $A \leq -1$; (b) B is real-symmetric; (c) $E := -\frac{r^2}{4}\Omega$ satisfies $AE = -E$, and E is the passive response.*

Proof. (a) $\mathcal{L}$ is self-adjoint on $L^2(G^{-1})$ with spectrum $\{-n/2\}$ (Hermite functions), and angular mode 2 first occurs at $n = 2$ [5]. (b) The multiplication part is clear. For the other part, $\Omega\rho = \frac{\Gamma}{4\pi}r$ exactly, so $\langle \Omega\Delta_2^{-1}X, Y \rangle_\rho = \frac{\Gamma}{4\pi} \int (\Delta_2^{-1}X)Y r dr$, which is symmetric. This is the mode-2 form of the skew-symmetry of Λ in [5, Lemma 2.3]. (c) is a computation. $\square$

Theorem 4.2. *Let $V = W - E$. For every $\Gamma > 0$, the first-order tilt satisfies*

$$\frac{T}{\lambda} \Big|_{\lambda=0} = -\pi^2 \int_0^\infty r^3 \text{Re} V dr \leq -\frac{16\pi^3}{\Gamma} \|V\|_{L^2(\rho)}^2 < 0.$$

Thus, to first order in λ , the vortex is tilted ahead and rounder than its passive twin at every circulation.

Proof. Since $AE = -E = \frac{r^2}{4}\Omega$, V solves $(A - iB)V = ib$ with $b = BE$ real. Pairing bilinearly with E and taking real parts, using $A = A^\top$ and $B = B^\top$, gives $\langle V_r, E \rangle = \langle V_i, b \rangle$. Pairing with $\bar{V}$ and taking real parts gives $\langle V_i, b \rangle = \langle AV, \bar{V} \rangle \leq -\|V\|^2$. Finally $E\rho = -\frac{\Gamma}{16\pi}r^3$, so the moment identity at first order reads $T/\lambda = -\pi^2 \int r^3 V_r = \frac{16\pi^3}{\Gamma} \langle V_r, E \rangle$. $V \neq 0$ because $b \neq 0$. Existence of V follows from the coercivity $\text{Re} \langle (A - iB)V, \bar{V} \rangle \leq -\|V\|^2$. $\square$

The sign of the forcing b is never used, nor any pointwise positivity. Three structural facts suffice: diffusion with strain is dissipative, self-induction is symmetric, and the passive response is an eigenfunction. Numerically (Section 6) the bound is nearly sharp: within 3% at $R = 1$, and to four digits for $R \geq 1000$. The tilt angle tends to the 45° of [11] as $R \rightarrow \infty$.

Proposition 4.3 (Time dependence). *Start from $\omega(0) = \Gamma G$. At first order in λ , V solves $\partial_t V = \mathcal{A}V - i(1 - e^{-t})b$ with $\mathcal{A} = A - iB$, $V(0) = 0$, and $T(t)/\lambda = \frac{16\pi^3}{\Gamma} \langle V_i(t), b \rangle$. Then $T(t) < 0$ for $0 < t < c\|b\|/\|\mathcal{A}b\|$ (of order $1/\Gamma$), and for all t with $(1+t)e^{-t}\|b\|^2 < \|V_\infty\|^2$, where $V_\infty = i\mathcal{A}^{-1}b$ (this holds for $t \gtrsim 2 \ln \Gamma$).*

Proof. The identity for $T(t)$ follows by pairing with E , as in Theorem 4.2, using the time-dependent moment identity. Early times: $V = -i\frac{t^2}{2}b + O(t^3\|\mathcal{A}b\|)$. Late times: $\|e^{\tau\mathcal{A}}\| \leq e^{-\tau}$ by dissipativity, so $\|V(t) - V_\infty\| \leq (1+t)e^{-t}\|b\|$; and $\langle V_{\infty,i}, b \rangle \leq -\|V_\infty\|^2$ by Theorem 4.2. Since $\|b\| \propto \Gamma^2$ and $\|V_\infty\| \propto \Gamma$, the threshold time is $2 \ln \Gamma + O(1)$. $\square$

5. LARGE CIRCULATION, EVERY ASYMMETRY

For every $\lambda \in [0, 1)$ and $\Gamma \geq R_0(\lambda)$ there is a steady vortex with $\|\omega - \Gamma G - \lambda w_\infty\|_{W^{1,2}(\infty;\lambda)} \leq \lambda C(\lambda)/(1 + \Gamma)$, where $\Lambda w_\infty = \mathcal{M}G$ and w_∞ is in modes ± 2 [9], [3, Thm. 4.2].

Theorem 5.1. *For every $\lambda \in (0, 1)$ there is $R_1(\lambda)$ such that, for $\Gamma \geq R_1$, this vortex satisfies $M_{11} = 2 + O_\lambda(\Gamma^{-1}) > 2/(1 + \lambda)$ and $T = -4\pi\Gamma\lambda + O_\lambda(1) < 0$.*

Proof. The mass is Γ , and w_∞ has zero mass. $|\int x_1^2 f| \leq C_\lambda \|f\|_{L^2(\infty;\lambda)}$. Insert this into $T = 4\pi\Gamma(1 - g_1 M_{11})$. $\square$

At large circulation the vortex is round to leading order [11], while its passive twin is not. The tilt is then forced by the moment identity.

Theorem 5.2. *Let $\phi = e^{-ax_1^2 - bx_2^2}$ with $0 < a < b$, and put $A = a + \frac{1}{4}$, $B = b + \frac{1}{4}$. For every $\lambda \in (0, 1)$ and $\Gamma \geq R_2(\lambda, a, b)$, the steady vortex of Theorem 5.1 satisfies*

$$\int \omega v \cdot \nabla \phi = \frac{\Gamma\lambda}{4} \int \phi (x_1^2 - x_2^2) G dx + O_{\lambda,\phi}(1) = \frac{\Gamma\lambda(b-a)}{32(AB)^{3/2}} + O(1) > 0.$$

Proof. By Proposition 2.6, $\int \omega v \cdot \nabla \phi = -\int \omega \mathcal{L}_\lambda^* \phi$. For the leading term, $\int G \mathcal{L}_\lambda^* \phi = \int \phi (\mathcal{L} + \lambda \mathcal{M})G = \lambda \int \phi \mathcal{M}G$, and $\mathcal{M}G = -\frac{1}{4}(x_1^2 - x_2^2)G$. The other terms are $O(\lambda)$ uniformly in Γ . The Gaussian integrals are explicit. $\square$

So at large circulation, positivity holds for the full Gaussian weight. It requires only that the weight is wider along the compressed axis, which the adjoint weights are (Proposition 2.3). For isotropic weights the leading term vanishes.

6. NUMERICAL EVIDENCE

All computations use the reduction (1) with $\nu = \gamma = 1$. Every computation was registered before it was run, with gates against exact solutions; the scripts and logs are kept with the author's working notes.

Monotonicity of the adjoint-weighted circulation (stochastic vortex method, 1200 blobs, from a round start). The self-induced term $\mathcal{S}$ is positive for 97–100% of the time in all twelve runs with $\lambda \in \{0.5, 1\}$, $R \in \{10, 30, 100\}$, and zero (to four digits) at $\lambda = 0$. Over $t \in [0, 10]$ the weighted circulation rises, for instance from 0.236 to 0.428 at $(\lambda, R) = (1, 100)$. Plain Gaussian weights instead lose up to half their value. The reduction of Proposition 2.4 tracks the full $\mathcal{S}$ with correlation 0.98. $T < 0$ for 99.6% of the time, and the moment identity of Proposition 2.2 holds to 2%.

Every circulation at first order (radial solver for (2), $R = 1$ –3000). $\text{Im } W \geq 0$ pointwise, and two independent evaluations of T agree to 3–4 digits. $T \propto R^3$ for weak vortices and $T \propto R$ for strong ones. The time-dependent problem gives $T(t) < 0$ for all $t \in (0, 14]$, also inside the window not covered by Proposition 4.3.

Intermediate circulation at finite asymmetry (finite-volume steady solver, 256^2). For $\lambda \in \{0.25, 0.5, 0.75\}$ and $R \in \{10, 30, 100, 300\}$, all twelve steady vortices are rounder than their passive twins, with $T < 0$; direct and identity-based values of T agree within 1.5%. $\int \omega v \cdot \nabla \phi > 0$ for six full Gaussian weights, including an isotropic one. At $R = 300$, T is within 3% of $-4\pi R\lambda$ (Theorem 5.1), and $\int \omega v \cdot \nabla \phi$ is within 1% of the leading term of Theorem 5.2.

What self-induction buys past $\lambda = 1$. Switching v off moves the survival boundary (the λ at which half the core circulation remains at $t = 10$) to $\lambda_c^{\text{pass}} = 1.07$, essentially $\lambda = 1$. That is where the intermediate eigenvalue of the strain changes sign. With self-induction the boundary moves to 1.07, 1.11, 1.36 and 2.22 for $R = 10, 30, 100$ and 300.

Lifetime in the torn regime. For $\lambda \in \{1.25, 1.5, 2, 3, 4\}$ and $R \in \{30, 100, 300, 1000\}$, from a round start, the half-life of the core circulation divided by the exact passive half-life is:

λ	$R = 30$	100	300	1000
1.25	1.08	2.83	> 5.3	> 5.3
1.5	1.05	1.79	> 8.2	> 8.2
2	1.02	1.32	> 13.7	> 13.7
3	1.01	1.12	4.29	> 24.4
4	1.01	1.06	2.21	$\geq 28^\dagger$

(“>” means the core was still intact at $t = 30$.) Ordered by $\varepsilon = 2\pi\lambda/R$, the twenty runs fall on one monotone curve. The gain is about 1 for $\varepsilon \gtrsim 0.25$; it rises through 1.3, 1.8, 2.8 and 4.3 as ε falls from 0.13 to 0.063; and every run with $\varepsilon \leq 0.042$ survives. The threshold is $\varepsilon_c \approx 0.05$. A sweep at $\lambda = 4$, $R = 400$ –700, registered with predictions read off this curve, confirmed it out of sample.

[†]At $\lambda = 4$, $R = 1000$ the core held 95% of its circulation and then left the measuring disc at $t \approx 24$, on domains of two sizes. This is not tearing. The vortex sits at the hyperbolic stagnation point of the strain; round-off displaces it, and the offset grows like $e^{|g_2|t}$ ($|g_2| = \frac{\lambda-1}{2}$). A run with an offset of 10^{-4} leaves at $t \approx 10$ for $\lambda = 3$, and with point symmetry enforced it stays at the origin, with its core intact.

Remark 6.1 (Positional instability). For the shape of the core the drift is harmless, since a linear strain is homogeneous. For a *collapse* it is not. A collapse takes place at the convergence point of its suppliers, which for $\lambda > 1$ is a hyperbolic stagnation point. An offset grows like $e^{(\lambda-1)\int s/2}$ while the core shrinks like $e^{-\int s/2}$, so the offset measured in core radii grows like $e^{(\lambda/2)\int s}$. Unless the suppliers follow the tube, a collapse fed in the tearing regime would have to be exactly symmetric.

7. SCALE INVARIANCE AND A COLLAPSING TUBE

Restore units. A tube in Burgers balance with axial rate s has core radius $a^2 = 4\nu/s$ and peak vorticity $\omega_0 = \Gamma/\pi a^2 = \Gamma s/4\pi\nu$. Its in-plane strain is $e = \lambda s/2$. Hence

$$\varepsilon = \frac{e}{\omega_0} = \frac{2\pi\lambda\nu}{\Gamma} = \frac{2\pi\lambda}{R},$$

independent of s . If a tube is compressed by an increasing axial strain, ε does not change. It depends only on the geometry of the strain (λ) and on the circulation (R). Both are invariant under the Navier–Stokes scaling. So the protection of a collapsing tube is decided at the critical level, the level at which the earlier papers of this series located the supercriticality gap [14, 15].

Three further facts constrain a hypothetical collapse. The strain a supplier induces on a tube has an in-plane part fixed by geometry: in a computation with ring-shaped suppliers, the ratio e/s is the same to within 1% for every kernel $K(r) \propto r|r|^{2\sigma-3}$, $\sigma \in [-\frac{1}{4}, \frac{1}{2}]$, of the

Euler–Arnold family (author’s notes). For a tube aligned with the most extensional direction, $\lambda > 1$ if and only if the intermediate eigenvalue λ_2 of the strain is positive. And by Miller’s criterion [10] enstrophy production, and hence any blowup, needs $\lambda_2^+ > 0$. A collapsing tube is therefore fed by strain in the tearing regime. To survive it, by the threshold of Section 6, it needs $R \gtrsim 2\pi\lambda/\varepsilon_c \approx 125\lambda$ at every scale of the collapse. By [15, Prop. 5.1] this means a persistently large value of $\|u\|_{\dot{B}_{\infty,\infty}^{-1}}/\nu$ at the collapsing scales. That is the quantity whose continuity in time decides regularity (Cheskidov–Shvydkoy [2]).

Remark 7.1 (What this does not give). Energy does not bound the circulation Reynolds number at small scales, since $L^2(\mathbb{R}^3)$ does not embed into $\dot{B}_{\infty,\infty}^{-1}$. Nothing here excludes persistently large R . The statements above *restate* the gap in terms of self-induction and do not close it. The theorems are proved on the two-dimensional reduction. Curvature of the tube and variation along its axis, the three-dimensional effects, are not treated. In the turbulence data we examined (the isotropic 1024^3 data set of the Johns Hopkins Turbulence Database [7]), intense tubes sit at $\lambda \approx 4$ (median) and $R \approx 250$ – 480 , just below the threshold $R_c \approx 500$: transient, which is consistent with regularity and proves nothing about it.

8. OPEN QUESTIONS

Question 8.1. *Is the steady asymmetric Burgers vortex rounder than its passive twin for every $\lambda \in (0, 1)$ and every Γ ? Uniqueness is unknown at intermediate Γ [3], so the statement would have to hold for every steady solution.*

Question 8.2. *Is $\int \omega v \cdot \nabla \phi \geq 0$ for the full adjoint Gaussian weight in general? For weak vortices this is the sign of one number, $\langle w_0, \sigma_\phi \rangle$, with $\sigma_\phi(y) = \int G_\lambda(x) K(x - y) \cdot (\nabla \phi(x) - \nabla \phi(y)) dx$. It is positive in every case computed, although σ_ϕ itself is not single-signed.*

Question 8.3. *Derive the threshold ε_c of the torn regime by averaging the strain over the fast rotation of the core. In the author’s computations it depends on the kernel ($\varepsilon_c \approx 0.03, 0.05, \gtrsim 0.08$ for $K \propto r^\perp |r|^{2\sigma-2}$, $\sigma = -\frac{1}{4}, 0, \frac{1}{4}$), through the radius at which the vortex’s own rotation falls to the strain.*

Question 8.4. *Does a weighted circulation of this kind remain monotone on a curved tube, where self-induction acquires the local-induction term $\frac{\Gamma \kappa}{4\pi} \ln(1/a\kappa)$ of Da Rios?*

ACKNOWLEDGEMENT

This paper was compiled with the help of generative AI. The author provided the direction: the tearing reading of external strain, the electromagnetic reading of self-induction, and the request for a heat-kernel monotonicity. The AI carried out the derivations and computations under the author’s supervision.

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ALMATY, KAZAKHSTAN

Email address: adilsalkimbayev@gmail.com